

ME 3910: Social/Community Project

Semester: 6

Priyangani Hotels and Bakery

Phase 03

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1. Introduction

The report is on the final phase of the Social/Community project conducted on Priyangani Hotel and Bakery which is a small and medium enterprise (SME) located in Kalagedihena, Sri Lanka. The phase 03 is focused on well-reasoned and structured solutions aligned with the Sri Lanka's National SME Policy Framework. The solutions are based on detailed background study and operational analysis in Phase 01 and from the identification of key challenges in Phase 02.



Figure 01: Priyangani Bakery Facility

During phases 01 and 02, various problems and challenges were identified based on the areas of SME development. The areas were legal status, outdated technology, skills development, financing limitations, market facilitation, research and development, resource efficiency, and regional outreach. These challenges affect the sustainable growth of Priyangani Hotel & Bakery. However, the majority of companies in Sri Lanka's food and hospitality industry deal with similar issues and difficulties, such as outmoded technology, trouble getting finance, a shortage of qualified personnel, and difficulties expanding into new areas.

In this phase strategic solutions will be proposed to problems in each identified SME area, according to the government policies, institutional support structures, and proven industrial practices related to food and bakery industry. Also, we studied where government and other related organizations that aren't providing sufficient support and suggested practical strategies with practical data measures and develop the effectiveness of the SME support framework on Priyangani Hotel & Bakery. Our aim is to provide effective and practical policies, solutions, and recommendations for the identified problems of Priyangani Hotel & Bakery, which will help the

business grow stronger and more successfully in this competitive industry. Additionally, these efforts will contribute to achieving national objectives related to SME development, job creation, and the economic growth of the country.

2. Solutions to Challenges Faced

2.1. Enabling Environment

Priyangani Bakers, like most small and medium-sized baking companies in Sri Lanka, is experiencing with a series of operational and regulatory challenges that limit its growth and competitiveness. The most noted challenge for us was the dominant business structure. As a sole proprietorship, the owner's personal property is exposed to the risks of business, unnecessarily putting them at risk of financial loss. Sole proprietorship also makes it difficult for Priyangani Bakers to obtain large bank loans, government funding schemes, and business associations that require limited liability status for qualification. The transformation of the business into a Private Limited Company (Pvt. Ltd.) will offer great legal and financial security, improved business credibility, and new paths for growth and formal associations.

Another operational issue we observed was the complex process of licensing. Today, Priyangani Bakers must obtain different municipal licenses separately for food safety, fire protection, waste management, and environmental clearance. This paperwork intensive, manual system increases the administrative expense to the operation of the business and results in delayed approvals, particularly when introducing new operations or products to the market. Moving towards a centralized licensing system under municipal government authority will significantly streamline the procedure.

Product standardization and quality control were the other areas of concern. Currently, there are no formal industry specific product quality, safety, and hygiene certifications for this bakery. This makes it difficult for the bakery to win customer confidence outside their currently operating area and this limits potential to supply to supermarkets, catering companies, hotels, restaurants or pursue export opportunities for products. It will improve consistency in products, safety, and marketability by implementing Sri Lanka accepted bakery standards and certification schemes through the Sri Lanka Standards Institution (SLSI) and training programs by the National Enterprise Development Authority (NEDA).

Fluctuating prices and inconsistent quality of raw ingredients like flour, margarine, yeast, and dairy products directly affect profitability and uniformity of the production. The Consumer Affairs

Authority (CAA) must have the power to manage pricing practices and quality problems by suppliers. Establishment of SME cooperative purchasing groups or procurement support programs at divisional or district levels can help Priyangani Bakers purchase high-quality raw materials at stabilized and fair prices.

Finally, there is policy advocacy. The government and industry associations need to come together and create supportive policies for Sri Lankan bakery companies, addressing concerns like tax policies, import duties on essential baking equipment, and promotional programs for Sri Lankan bakery brands.



Figure 02: Relevant Government Authorities

2.2. Modern Appropriate Technology

Priyangani Bakers currently employs hand-planned manufacturing schedules across production and outdated bakery equipment, contributing to inefficiencies in operations, higher labor costs, and greater food wastage. According to the bakery owners, about 8% to 10% of excess bakery items are produced. The absence of modern machinery like automated dough divider rounders, digital proofing cabinets, and programmable convection ovens is likely to make the production time-consuming and inconsistent, specially at peak demand periods. This also limits the scope and quality of products that can be offered consistently.



Figure 03: Digital Proofing Cabinets (Left) and Automated Dough Rounders (Right)

To fill up this gap, the bakery can be improved by obtaining financial assistance under such schemes as the National Science Foundation's Technology Transfer and Development Fund (TTDF). Equipment upgrade strategy would improve productivity, conserve waste, and reduce energy consumption, thereby improving the business to make it more competitive and sustainable in the long term.

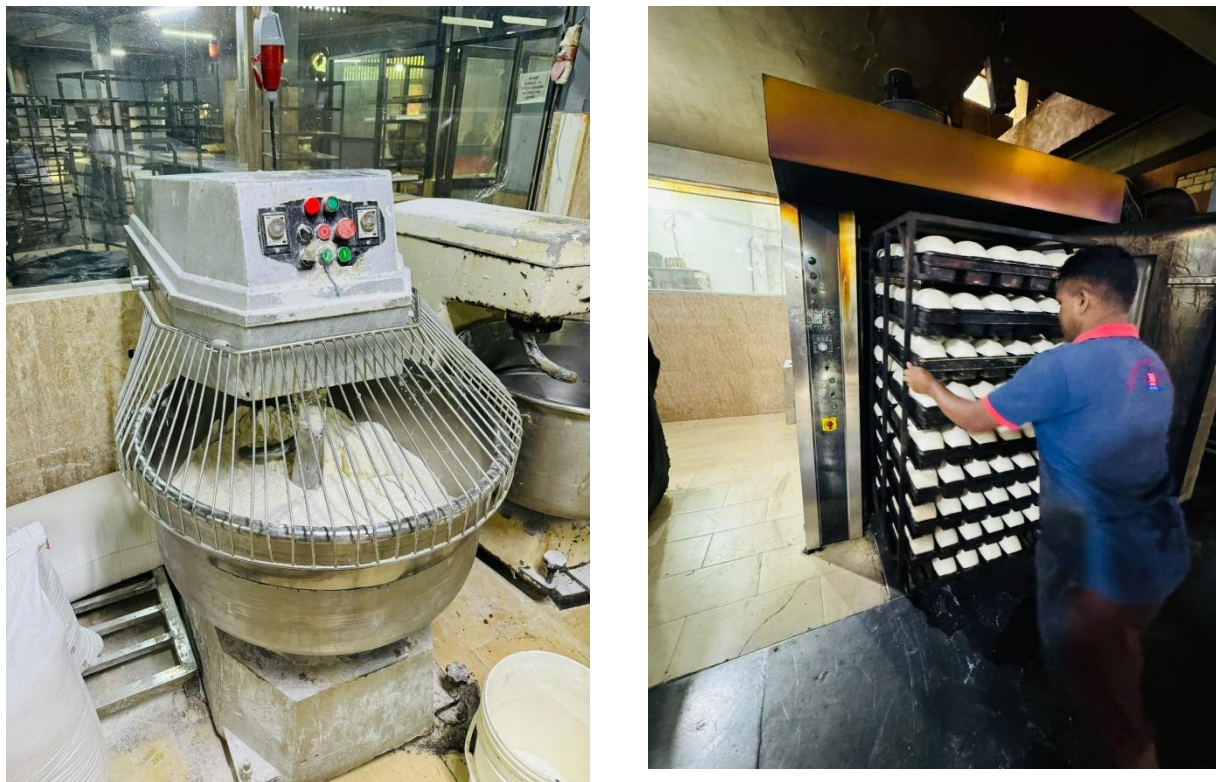


Figure 04: Currently Used Dough mixing machine (Left) and Oven (Right)

Now, Priyangani Bakers manually conduct quality control checks with whiteboards and marker notes to track quantities produced each day, batches of ingredients, and quality of products. Daily demand is forecasted using the demand of the previous day. This method has a greater possibility of error, is not traceable, and cannot detect production issues in real time. Instituting a Point of Sale (POS) system in addition to an inventory management application would allow the bakery to monitor electronic sales trends, inventories, and customer preferences, ensuring production planning on data. This also enables the opportunity to forecast future demands with more reliable mathematical methods without using old fashioned techniques. This kind of system would reduce material wastage, avoid stockouts, and improve order fill rates. It would also simplify financial reporting as well as regulatory compliance by generating computerized reports for tax submissions as well as business analysis.

Additionally, the lack of technology tools for running management delays Priyangani Bakers' ability to scale. Many modern bakeries are capitalizing on software packages for automating order handling, ingredient ordering, and scheduling from customers. The government can also introduce discounted packages for small food manufacturers when purchasing software at affordable prices. Technical colleges and vocation training schools can offer bakery staff practical exposure to these applications.

2.3 Entrepreneurial Culture and Skills Development

One of the most emerging problems that we found out was the aging workforce, with most of long-term employees having acquired their expertise informally using experience based learning as opposed to formal education. The informal passing on of knowledge from one generation to the other generation has sustained baking operation for decades and it threatens consistency, productivity, and adaptation to modern, easier and reliable ways. Without formal procedures for training, operating skills are lost when experienced personnel retire or leave. Another main issue that the owner brought up is, the younger generation has a growing uninterest towards conventional bakery work. Most of them prefer modern, technology driven workplaces with sleek career progression, training, and a clean working environment. This poses serious threats to long term sustainability of the bakery business. Because of that, most of young people are not willing to pursue these professions.

To overcome this, experienced professionals in the industry could organize structured induction programs and practical training sessions to properly guide new workers. These programs should include comprehensive orientation courses on food safety legislation, standard operating procedures, use of modern baking machines, and customer handling skills. Once those processes are properly standardized, the bakery can deliver consistent quality and run operations smoothly, even with the continuously changing workforce from time to time. It would also be helpful to work with institutions like SLITHM and NAITA to create well-structured training programs for new recruits, ensuring they learn the right skills from the start. Currently SLITHM offers certificate courses regarding the bakery industry. Through this collaboration, Priyangani Bakers would be able to engage with new trainees through certification programs, equipping them with practice-based, industry-specific skills while creating a stable source of quality young people.

On the other hand, to bring young workers into the organization and establish an innovative culture, Priyangani Bakers can conduct weekly innovation sessions in which employees can bring in their own ideas for new recipes, new food items, new methods of production, packaging solutions, or business improvement. Bakery industry is a traditionally male dominant field. By introducing women-oriented training programs and internships in association with vocational training centers, more female participation in the baking industry can be expected. Through flexible schedules of work, training programs, and friendly working environments for women, Priyangani Bakers can make use of previously untapped portion of the labor pool, while also boosting social inclusion and business development.

2.4 Access to Finance

Priyangani Hotel & Bakery is currently self-funded and has faced challenges in securing outside financial support. This is mainly because business doesn't have valuable assets to offer as security and lacks proper financial planning. The business runs under sole proprietorship status. This limits access to bank loans and proper financial planning. And also, they haven't been able to get loans because the business isn't office registered as a private company.

To address these issues, the bakery should be encouraged to re-register as a (Pvt.) Ltd. company. This will improve credibility and make it eligible for SME focused financial programs. Additionally, offering non collateral based credit options and government-backed loan guarantees

can help such businesses secure the necessary capital. These credit programs focus more on repayment capacity of the company and the probability of success of the program.

Support initiatives led by government like tax breaks, low-interest SME loans, or Central Bank funding programs like the Prosperity Loan Scheme can play a key role in growing the company and upgrading its technology. Furthermore, collaborating with organizations such as the National Enterprise Development Authority (NEDA) and financial advisory agencies can help the company create a solid financial plan and find the correct sources of funding.

Enhancing the internal record-keeping of the company can also enhance its chances of receiving funding. Tracking expenses, maintaining accurate income records, and producing simple financial statements can all help show lenders how well a business is performing and how capable it is of repaying debt. Participating in workshops or SME targeted financial training programs run by organizations like NEDA or local chambers of commerce can also be beneficial for the bakery. In addition to facilitating financial access, these initiatives will enhance long-term sustainability and overall financial management.

2.5. Market Facilitation

Priyangani Hotel & Bakery faces significant challenges in expanding its market reach. Branding remains underdeveloped, packaging is basic, and there is virtually no online presence. Additionally, the business isn't connected with delivery platforms and modern sales channels. So it limits the market outreach and customer engagement.

To strengthen market facilitation, the company should focus on rebranding. This includes updating the logo, improving the packaging design, and creating a professional website that highlights the product range, business story, and allows customers to place orders easily. Building an active presence on social media platforms like Facebook and Instagram through regular, creative content and customer engagement will significantly improve visibility.

Participation in food expos, trade shows and SME marketing programs led by the government will also provide chances to build connections and expand customer bases. Government agencies like NEDA (National Enterprise Development Authority) and the SME Development Division can facilitate these opportunities and monitor the bakery's market growth.

Additionally, integrating with delivery platforms and e-commerce channels will allow the bakery to meet modern customer preferences and access new markets efficiently. Online ordering and home delivery have become essential services for many customers, especially in urban areas. By joining well known delivery apps like uber eats pick me and setting up dedicated online store, the bakery can reach a broader customer base, increase sales, and compete more effectively in today's digital-driven market.



Figure 05: Small Enterprises Development Division

2.6. Research and Development

Priyangani Hotel & Bakery currently lacks a proper research and development (R&D) process in place. Most of the decisions related to new products and recipes are made based on the experience of the staff rather than proper testing and research. There is no proper way to test how long products stay fresh and ensure food safety guidelines are met. And there is no proper way to begin new and more efficient baking methods. As a result, the bakery finds it difficult to maintain consistent product quality, minimize waste and stay competitive with competitors who use more updated methods. This limits the bakery's ability to grow, improve its products, and meet the changing customer demands.

Collaborating with nearby universities and research centers can offer important technical support for product development and quality testing of the products. Such collaborations could help determine optimal storage conditions or improve the health value of baked products. Government institutions like NEDA can also act as facilitators in connecting small businesses with suitable research partners.

Furthermore, Priyangani Bakery should also consider setting up a small research and development team responsible for introducing new recipes, improving food safety standards, and dealing with new energy saving cooking methods. And allocating a specific budget and time for the R & D activities is important to ensure they operate smoothly without disrupting the bakery's daily workflow. Support from the government through grants for innovation and technical consultancy services can further boost these R & D efforts of the Priyangani bakery, allowing the business to remain competitive and compliant with local and international food standards.

Finally, Priyangani bakery should organize regular internal meetups to discuss R & D outcomes with the production team. Tracking key factors like product quality, waste reduction, and customer feedback, the bakery can gauge the impact of its R&D initiatives and make data-driven decisions for future growth.

2.7 Efficiency in Resources

Remarkable waste-reduction measures have been taken by Priyangani Bakery, such as turning unsold bread into biscuit powder and giving stale buns to animals as feed. High-energy animal feed is created by this repurposing process and is frequently fed to pigs and poultry. However, there are still issues in important areas like packaging and energy use. In example, the Kalagedihena shop still uses diesel-powered ovens and uses more than 6,200 kWh of electricity on average each month. Furthermore, using single-use plastic boxes for delivery adds to the contamination of the environment and needless plastic trash.

A number of actions are suggested to remedy these inefficiencies and lessen their impact on the environment. First off, fuel consumption and CO₂ emissions may be greatly reduced by switching from diesel ovens to LPG or electric ones, ideally with heat recovery systems. With built-in heat exchangers, contemporary bakery ovens may recover waste heat from exhaust gasses to warm water or air, saving a significant amount of energy. Second, by constructing a rooftop solar photovoltaic (PV) system with a capacity of around 10 kW, the bakery may stabilize energy prices and lessen its dependency on fossil fuels by offsetting 10–20% of its grid power use. Thirdly, the environmental impact of single-use plastic packaging may be reduced by switching to reusable HDPE bread boxes, which are made to be strong, stackable, and simple to clean. Last but not least, the bakery would be able to systematically monitor resource

consumption, detect inefficiencies, and confirm the efficacy of made changes by implementing monthly water and energy audits based on the ISO 50001 methodology.

By doing these things, Priyangani Bakery may save money, increase operational efficiency, and promote environmental sustainability, all of which will benefit the company and the society at large.

2.8 Regional Balance

At present, the four Priyangani Bakery stores are located within open places within the urban limits of Gampaha District and outlying Divisional Secretaries (DS) divisions including Mirigama and Veyangoda lack quality baked goods. This leaves an imbalance of distribution of services in the region. On the supply side, Priyangani Bakery has already been using in-country sources to acquire some of key sources such as eggs and coconut oil, this is important as it helps the local economy. By buying in local smallholder farmers, especially those that control most of the sectors such as coconut farming in Sri Lanka, not only does it ensure the livelihoods in the rural areas but also supports the agricultural chain and well-being of many communities.

To further expand the social and economic opportunities to unserved rural regions. These are the local bakery units or outlets owned at small scale but functioning in the Priyangani brand can serve and deliver quality products to the far-flung places and worked on local basis offering local jobs. Working together with regional Chambers of Commerce (Gampaha or Mirigama) would allow specific training to potential franchisees in such fields like bakery operation, accounting and marketing, turning them into entrepreneurs and grass roots development. Also, the continuity and growth of providing key ingredients such as eggs and coconut-oil around 100 percent of locally based farm will not only retain the monetary value of the economy in the society, but will also minimize transportation emissions as well as coining freshness via farm to table.

Such strategic interventions do not only enhance the availability of good food in rural places, but also catalyse inclusive economic growth by empowering the local food systems, facilitating smallholder farmers, and reinforcing community resilience, at the regional level.

2.9 Digital Transformation & Governance

Within Priyangani Bakery, their non-production records such as accounting, payrolls, human resources, and maintenance records are run on hand written ledgers. This antiquated system is greatly harming the efficiency of their operations and control over their management since the information retrieval process is cumbersome and it is hard to trace the trends of their finances or follow the maintenance of the equipment. By shifting to digital tools, there is a bright chance to revise the operations and foster the governance practices. The bakery can automate processes like the generation of invoices, expenses tracking, and salary management by using a cloud-based accounting and payroll tool and joining cloud-based accounting, such as QuickBooks Online. It not only minimizes the incidence of human errors and duplication of costs and efforts but also real-time financial reporting available in any device. Such characteristics as backups in real time and safe access guarantee data security, enhancing the work transparency and decision-making process at the same time.

Moreover, the use of QR code-powered maintenance records of machines and cars would enable employees to follow the record of repairs and future maintenance online, enabling the improved control over assets and minimizing downtimes. The bakery must also create quarterly board-style review meetings to reinforce a good governance, despite being a sole-directorship entity, after the formal registration of the company. This kind of systematic meetings would help in sustaining accountability, promote long range planning and the establishment of a documented framework within which the business performance is assessed and sustainable growth steered.

3. Conclusion

The report outlines practical and effective solutions to the challenges faced by Priyangani Hotel & Bakery. However, this describes a broader context of solutions faced by most of the SME hotels and bakeries in Sri Lanka.

The solutions were developed with reference to the National SME Policy Framework where it mainly focuses on legal status, outdated technology, workforce development, financial access, market expansion, research and development, resource efficiency, regional balance, and digital transformation of the bakery.

The proposed policies and strategies will be helpful for the company to improve the effectiveness of the production and quality of their pastry and bakery products in this competitive market. Therefore, the solutions formal business registration, upgrading to modern equipment, adopting digital management systems, structured staff training, financial literacy improvements, market rebranding, and renewable energy utilization will solve most of the challenges identified on the SME framework.

Also, encouraging government bodies and similar organizations who can be helpful in the development of enterprises like Priyangani Hotel & Bakery will enhance the SME framework as a policy mechanism on such enterprises in Sri Lanka.

These recommendations will contribute not only to the long-term growth of individual businesses but also to achieving Sri Lanka's broader goals of economic development, new job opportunities, and quality product output.

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